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Network Availability Monitoring and Packet Loss Analysis Using Internet Control Message Protocol

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
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BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Network Availability Monitoring and Packet Loss Analysis Using Internet Control Message Protocol” has been carried out by P Surendra Kumar Reddy(192425224),B Rajesh(192472165),L Keshavaram (192511253) & S V Harish Kumar (192511486) under the supervision of DR.P. JESU JAYARIN and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
P Surendra Kumar Reddy (192425224)	Network architecture, ICMP concepts, requirement analysis	Designed the network monitoring framework and ICMP-based connectivity mechanism	Tested network reachability and analyzed ICMP response behavior	Contributed to Introduction, Objectives, and Technical Foundation	25%
B Rajesh (192472165)	Packet loss monitoring, data collection, performance analysis	Developed the packet loss detection and measurement methodology	Conducted packet loss, latency, and response-time analysis	Contributed to Methodology, Packet Loss Analysis, and Results	25%
L Keshavaram (192511253)	System implementation, monitoring workflow, data visualization	Developed the monitoring workflow and implemented ICMP request-response processing	Tested different network conditions and verified monitoring accuracy	Contributed to System Design, Implementation, and Discussion	25%
S V Harish Kumar (192511486)	Documentation, validation, security considerations, overall integration	Assisted in integrating the monitoring components and improving system functionality	Performed final validation, troubleshooting, and result verification	Contributed to Conclusion, Applications, Limitations, and References	25%
Total					100%

ABSTRACT

Network availability is an essential requirement for reliable communication in modern computer networks, where connectivity failures, packet loss, and delayed responses can significantly affect network performance and user services. Conventional troubleshooting methods often depend on manual ping tests and may not provide continuous monitoring, systematic packet-loss analysis, or clear availability measurements. Therefore, there is a need for a simple, reliable, and cost-effective engineering solution that can continuously evaluate network connectivity and identify potential communication problems.

This project, titled “**Network Availability Monitoring and Packet Loss Analysis Using Internet Control Message Message Protocol,**” proposes an ICMP-based network monitoring system for detecting network availability and analysing packet loss. The system uses ICMP Echo Request and Echo Reply messages to periodically test the connectivity of selected network hosts. The proposed solution records successful and failed responses and calculates network availability and packet-loss percentages. Python is used to implement the monitoring logic, while Cisco Packet Tracer is used to create and test the network environment. Wireshark is used to capture and analyse ICMP packets to verify the communication process and monitoring results.

The methodology consists of requirement analysis, system architecture design, ICMP monitoring implementation, network simulation, packet capture, experimental testing, and performance evaluation. Multiple network conditions were considered, including normal connectivity, low packet loss, moderate packet loss, high packet loss, and completely unreachable hosts. The experimental results demonstrated that the system can correctly identify connectivity conditions and calculate packet loss and availability. Under normal conditions, 100% availability and 0% packet loss were observed. A 5% packet-loss condition resulted in 95% availability, while a completely unreachable host produced 100% packet loss and 0% availability.

The results demonstrate that ICMP-based monitoring provides a lightweight and practical method for basic network availability assessment and packet-loss analysis. The proposed system can assist network administrators in identifying connectivity degradation and supporting troubleshooting activities. However, ICMP responses may be affected by firewalls, rate limiting, congestion, or security policies. Future improvements can include real-time dashboards, historical data storage, automated alerts, latency analysis, and intelligent anomaly detection.

Keywords: ICMP, Network Availability, Packet Loss, Network Monitoring, Wireshark, Python

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
ICMP	Internet Control Message Protocol	—
IP	Internet Protocol	—
IPv4	Internet Protocol Version 4	—
TCP/IP	Transmission Control Protocol / Internet Protocol	—
TCP	Transmission Control Protocol	—
UDP	User Datagram Protocol	—
RTT	Round-Trip Time	ms
IETF	Internet Engineering Task Force	—
RFC	Request for Comments	—
SNMP	Simple Network Management Protocol	—
IPPM	IP Performance Measurement	—
ICN	Information-Centric Networking	—
IDS/IPS	Intrusion Detection System / Intrusion Prevention System	—
SMS	Short Message Service	—
PL	Packet Loss	%
A	Network Availability	%

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Network availability and reliable data communication are essential for modern organizations, educational institutions, cloud services, and enterprise applications. A network may experience connectivity failures, high latency, congestion, or packet loss due to faulty network devices, overloaded links, transmission errors, configuration problems, or unstable connections. Internet Control Message Protocol (ICMP) is a network-layer protocol commonly used for diagnosing and monitoring IP-based networks. Tools such as ping and traceroute use ICMP messages to determine whether a destination is reachable and to obtain information about network performance.

This project, “Network Availability Monitoring and Packet Loss Analysis Using Internet Control Message Protocol,” focuses on developing a monitoring approach that uses ICMP Echo Request and Echo Reply messages to evaluate network availability and packet delivery performance. By periodically sending ICMP packets to selected network devices and analyzing the responses, the system can identify unreachable hosts, calculate packet-loss percentage, and observe variations in network connectivity. The proposed solution provides a simple and practical method for detecting network problems and supporting network administrators in troubleshooting connectivity issues.

1.2 Need for the Project

Modern networks are expected to provide continuous and reliable connectivity. Even short periods of network unavailability or packet loss can affect web applications, online classes, cloud services, video conferencing, file transfers, and other network-dependent operations.

The project is needed for the following reasons:

- Problem identification: Network failures and packet loss need to be detected quickly before they significantly affect users and services.
- Affected users: Network administrators, organizations, students, employees, service providers, and end users can be affected by poor network availability.
- Existing limitations: Manual network testing is time-consuming, while basic connectivity tools may provide limited information about packet-loss behavior and network reliability over time.
- Engineering significance: Automated ICMP-based monitoring provides a cost-effective

approach for continuously measuring reachability and packet delivery, helping engineers identify potential network faults and performance degradation.

The project therefore provides an engineering-oriented solution for monitoring network health and supporting systematic troubleshooting.

1.3 Problem Statement

Network connectivity can be affected by multiple interacting factors, including device failures, link congestion, transmission errors, routing problems, configuration errors, and unstable network conditions. These factors may result in intermittent connectivity and packet loss that are difficult to identify through occasional manual testing. At the same time, a monitoring system must provide timely detection while minimizing unnecessary network traffic and processing overhead.

Therefore, the engineering problem is to design and implement an ICMP-based network monitoring system capable of continuously determining host availability, measuring packet loss, and analyzing connectivity behavior under realistic network conditions, while maintaining low monitoring overhead and providing useful information for network troubleshooting.

1.4 Project Aim

The aim of this project is to design and implement an ICMP-based network availability monitoring and packet-loss analysis system that continuously evaluates network connectivity, identifies unreachable hosts, measures packet loss, and supports efficient network fault detection and troubleshooting.

1.5 Project Objectives

The main objectives of the project are:

1. To design an ICMP-based framework for monitoring the availability of network hosts.
2. To develop a mechanism for sending ICMP Echo Request packets and receiving Echo Reply responses.
3. To model network availability and packet-loss conditions using appropriate monitoring parameters.
4. To implement automatic calculation of packet-loss percentage and host availability.
5. To test the monitoring system under normal, delayed, and packet-loss network conditions.
6. To evaluate the collected results to identify connectivity problems and network performance degradation.

7. To provide a simple monitoring interface or reporting mechanism for presenting network status and analysis results.

1.6 Scope

The project includes:

- ICMP Echo Request and Echo Reply-based connectivity monitoring.
- Monitoring of selected IPv4 network hosts/devices.
- Detection of reachable and unreachable hosts.
- Measurement and calculation of packet-loss percentage.
- Analysis of successful and failed ICMP requests.
- Basic monitoring of response behavior and connectivity status.
- Logging or displaying monitoring results.
- Testing using a simulated or controlled network environment.
- Generation of useful information for network troubleshooting.

Excluded

The project does not primarily cover:

- Deep packet inspection of application-layer traffic.
- Full intrusion detection or prevention.
- Detailed analysis of TCP, UDP, or application-layer protocols.
- Automatic configuration of routers and switches.
- Complete enterprise network management functionality.
- Physical repair of network devices or transmission media.
- Guaranteed detection of every possible network fault.

Assumptions

- The monitored devices support IPv4 and ICMP communication.
- Network devices are configured with valid IP addresses.
- ICMP traffic is permitted by the network or firewall configuration.
- The monitoring system has network connectivity to the monitored hosts.
- Test results are collected under controlled or known network conditions.

Boundary Conditions

The accuracy of monitoring depends on factors such as ICMP filtering, firewall rules, network

congestion, routing conditions, and the selected monitoring interval. An ICMP timeout does not always indicate complete network failure because a device may intentionally block or deprioritize ICMP traffic. Therefore, the system primarily evaluates ICMP-based reachability and packet delivery, rather than guaranteeing complete application availability.

1.7 Expected Engineering Outcome

The expected outcome of the project is a functional prototype of an ICMP-based network monitoring system capable of:

- Monitoring the availability of selected network hosts.
- Sending and receiving ICMP packets automatically.
- Identifying successful and failed connectivity attempts.
- Calculating packet-loss percentage.
- Displaying network availability and packet-loss results.
- Maintaining monitoring records for analysis.
- Providing useful information for network fault detection and troubleshooting.

The final solution is expected to demonstrate how ICMP can be practically applied as a lightweight network monitoring mechanism. It will provide an engineering model that can be further extended with features such as real-time dashboards, alerts, historical analysis, latency monitoring, and graphical visualization.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review for this project was conducted by examining relevant **Internet Engineering Task Force (IETF) standards, technical documentation, research literature, network-monitoring technologies, and commercial monitoring systems** related to ICMP, network availability, packet loss, latency, and connectivity analysis. Particular attention was given to **RFC 792**, which defines the Internet Control Message Protocol, and **RFC 1122**, which specifies requirements for Internet hosts and describes ICMP functionality.

Existing network-monitoring solutions such as **Nagios, Zabbix, and PRTG Network Monitor** were also reviewed to understand how ICMP Echo Requests, Echo Replies, response time, packet loss, thresholds, and alerts are used in practical network-management environments. The review focused on identifying the advantages, limitations, implementation approaches, and engineering requirements of existing solutions and determining opportunities for a simpler ICMP-based monitoring and packet-loss analysis system.

2.2 Review of Existing Work

2.2.1 Internet Control Message Protocol

ICMP is an important component of IP networking that provides diagnostic, error-reporting, and control functionality. RFC 792 defines ICMP messages and their operation within IP networks, while RFC 1122 specifies requirements for ICMP implementation in Internet hosts. ICMP Echo Request and Echo Reply messages form the fundamental mechanism behind the commonly used ping utility. For network monitoring, an ICMP Echo Request is transmitted to a destination host. If the destination is reachable and permits ICMP responses, an Echo Reply is returned. The monitoring system can use the response to determine availability and calculate response time. Repeated requests can additionally be used to estimate packet loss.

Relevance: ICMP provides a lightweight and standardized foundation for the proposed monitoring system.

2.2.2 Ping-Based Network Monitoring

Ping is one of the simplest methods for testing network reachability. It sends ICMP Echo Requests and waits for corresponding Echo Replies. The results can be used to determine whether a host is reachable and to measure round-trip time and packet loss.

The major advantage of ping is its simplicity and low implementation complexity. However, a basic ping command normally provides information for a limited observation period and does not necessarily provide continuous historical monitoring, centralized analysis, or customized reporting.

Relevance: The proposed project extends the basic ping concept into a structured monitoring system that continuously records availability and packet-loss information.

2.2.3 Nagios

Nagios is a widely used network and infrastructure monitoring platform. Its `check_ping` functionality uses ICMP to determine whether a network device is available. It can also use configurable warning and critical thresholds for response time and packet loss. Nagios documentation demonstrates that monitoring thresholds can be adjusted according to the characteristics of different devices and networks.

The major advantage of this approach is its ability to integrate ICMP checks into a larger monitoring and alerting framework. However, configuring and managing a complete monitoring platform can introduce additional complexity when the requirement is limited to basic availability and packet-loss analysis.

Relevance: Nagios demonstrates the practical usefulness of ICMP monitoring and threshold-based fault detection.

2.2.4 Zabbix ICMP Monitoring

Zabbix provides an ICMP Ping monitoring template that collects **ICMP availability, ICMP packet loss, and ICMP response time**. It also provides triggers for conditions such as repeated timeouts, high packet loss, and excessive response time.

This approach provides more comprehensive monitoring than a basic ping utility because it combines measurement, thresholds, and automated status evaluation. However, Zabbix is a large-scale monitoring platform containing many features beyond the requirements of a focused academic ICMP monitoring project.

Relevance: The Zabbix approach provides useful concepts for defining monitoring parameters and alert conditions in the proposed system.

2.2.5 PRTG Network Monitor

PRTG provides a dedicated Ping Sensor that uses ICMP Echo Requests to monitor device availability. Its implementation can report ping time, minimum and maximum response time, and packet loss when multiple ICMP requests are sent during a monitoring interval.

An important feature is the use of multiple ping requests within an interval, which allows the system to calculate packet loss rather than simply reporting whether a single ping succeeded. PRTG also provides configurable monitoring behavior and visualization capabilities.

Relevance: The concept of repeated ICMP requests and packet-loss calculation is directly applicable to the proposed system.

2.2.6 Commercial and Existing Monitoring Technologies

Existing network-monitoring platforms generally combine several technologies, including ICMP, TCP/UDP checks, SNMP, dashboards, logging, threshold analysis, and alert mechanisms. These systems are designed for large-scale infrastructure monitoring and therefore provide extensive functionality.

However, this functionality can also make them more complex than necessary for a focused project whose primary objective is to **measure network availability and packet loss using ICMP**. The proposed work therefore concentrates on the core ICMP monitoring mechanism while maintaining measurable performance indicators and clear analysis.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Basic Ping / ICMP Echo	Simple, lightweight, easy to use, directly tests reachability	Limited historical analysis and automation	Provides the fundamental ICMP mechanism
Nagios ICMP Monitoring	Supports availability checks, packet-loss thresholds, response-time thresholds and alerts	Requires monitoring-platform configuration; broader than the project requirement	Provides concepts for threshold-based monitoring
Zabbix ICMP Monitoring	Measures availability, packet loss and response time; supports triggers	Large and relatively complex monitoring platform	Provides a model for automated ICMP measurement and status evaluation
PRTG Ping Sensor	Measures availability, response time and packet loss using repeated ICMP requests	Commercial platform with extensive features beyond the project scope	Provides a reference for repeated-packet analysis
Manual Network Testing	Requires minimal setup and is useful for quick troubleshooting	Time-consuming, inconsistent and unsuitable for continuous monitoring	Demonstrates the need for automation
Proposed ICMP	Focused, customizable, lightweight and suitable for	Primarily limited to ICMP-based network	Integrates availability monitoring and

Monitoring System	academic/controlled environments	observations	packet-loss analysis into a focused solution
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The comparison shows that existing commercial platforms already provide sophisticated ICMP monitoring capabilities. However, the proposed project focuses specifically on understanding and implementing the underlying engineering process rather than depending on a complete enterprise monitoring platform.

2.4 Research / Engineering Gap

The review indicates that ICMP-based monitoring is well established and is already incorporated into several professional network-monitoring platforms. Existing systems can measure availability, packet loss, and response time effectively. However, they are generally designed as **large-scale infrastructure-monitoring platforms**, containing numerous features that may not be necessary for a focused educational or prototype implementation.

A key engineering gap is therefore the development of a **simple, transparent, and customizable ICMP monitoring solution** that demonstrates the complete process from ICMP packet transmission to result collection, packet-loss calculation, availability determination, and performance analysis.

Another limitation is that an ICMP failure cannot always be interpreted as complete service failure because network devices or firewalls may filter, restrict, or deprioritize ICMP traffic. RFC 1122 also notes that certain ICMP error conditions should be interpreted as indications rather than absolute proof of an unreachable destination. Therefore, the proposed system should clearly distinguish **ICMP reachability** from complete application availability.

2.5 Proposed Contribution

The proposed project contributes a **focused ICMP-based network availability and packet-loss monitoring system** designed specifically for practical analysis and troubleshooting. Instead of relying on a complete enterprise monitoring platform, the project concentrates on implementing the essential monitoring functions in a transparent and understandable manner.

The proposed contribution includes:

1. **Automated ICMP monitoring** – Periodically sends ICMP Echo Requests to selected hosts.
2. **Availability detection** – Determines whether monitored hosts respond successfully.
3. **Packet-loss analysis** – Calculates packet-loss percentage from successful and unsuccessful requests.
4. **Response analysis** – Records response behavior to identify connectivity degradation.
5. **Threshold-based evaluation** – Identifies abnormal packet-loss or connectivity conditions.
6. **Result visualization/reporting** – Presents monitoring results in an understandable format.

7. **Troubleshooting support** – Provides information that can assist in identifying possible connectivity problems.
8. **Lightweight architecture** – Avoids unnecessary enterprise-level complexity while retaining the essential functionality required for network availability analysis.

Overall, the project builds upon the established ICMP Echo mechanism defined in networking standards and the monitoring approaches demonstrated by systems such as Nagios, Zabbix, and PRTG, while developing a **simplified and customizable prototype focused specifically on network availability and packet-loss analysis**.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

3.1.1 Stakeholder / User Requirements

Stakeholder	Requirement
Network Administrator	Monitor the availability of network devices and hosts continuously.
Network Administrator	Identify packet loss and connectivity failures quickly.
System Operator	View monitoring results in a simple and understandable format.
Organization	Obtain reliable network-performance information with low monitoring overhead.
End User	Experience improved network reliability through early fault detection.
Project Developer	Implement a lightweight, configurable and maintainable monitoring system.

3.1.2 Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Send ICMP Echo Requests	Functional	Successfully send requests at configured intervals
Receive and process ICMP responses	Functional	Record successful/failed responses
Detect host availability	Functional	Display reachable/unreachable status
Calculate packet loss	Functional	Calculate loss percentage correctly
Record response information	Functional	Store results for each monitoring cycle
Generate monitoring report	Functional	Display summarized monitoring results
Configure target hosts	Functional	Support multiple user-defined IP addresses
Monitoring interval	Non-Functional	Configurable, e.g., 1–60 seconds
Packet-loss calculation accuracy	Non-Functional	Correctly calculate percentage from test results
System response	Non-Functional	Results displayed within a reasonable monitoring interval

Reliability	Non-Functional	Continue monitoring during temporary packet loss
Usability	Non-Functional	Simple interface and clear status indicators
Security	Non-Functional	Restrict monitoring configuration to authorized users where applicable

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
RFC 792 – ICMP	Defines ICMP message formats and functions	Used as the technical foundation for Echo Request and Echo Reply monitoring
RFC 1122 – Internet Host Requirements	Defines requirements for Internet hosts and ICMP behavior	Used to understand correct ICMP-based host communication
IPv4 (RFC 791)	Defines IPv4 packet addressing and delivery	Used for identifying and communicating with monitored IPv4 hosts
TCP/IP Architecture	Provides the networking framework for IP communication	Used as the underlying communication architecture
Operating System Network Security Policies	Network operations should comply with local security policies	ICMP monitoring is restricted to authorized/controlled hosts
Data Protection Principles	Avoid unnecessary collection of sensitive information	The system focuses on network-performance information rather than user data

Major Constraints

- ICMP may be blocked by firewalls or network security policies.
- Packet loss can be caused by temporary congestion and may not always indicate device failure.
- Monitoring accuracy depends on network conditions.
- Excessive ICMP requests can create unnecessary network traffic.
- The prototype may not represent the complete functionality of enterprise monitoring.

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
Monitoring protocol	ICMP Echo Request/Reply	—	High
IP version	IPv4	—	High
Monitoring interval	1–60	seconds	High
Packet count	Configurable	packets/cycle	High
Packet-loss calculation	$(\text{Failed Packets} / \text{Sent Packets}) \times 100$	%	High
Availability status	Reachable / Unreachable	—	High
Response monitoring	Record ICMP response	ms	Medium
Target hosts	User-configurable	Hosts	High
Monitoring mode	Continuous/periodic	—	High
Data logging	Store monitoring results	Records	Medium
Alert threshold	Configurable packet-loss limit	%	Medium
Network overhead	Minimized ICMP traffic	—	High
Interface	Simple monitoring dashboard/report	—	Medium
Security	Authorized monitoring only	—	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Basic Ping-Based Monitoring

A simple command-line implementation periodically sends ICMP Echo Requests to selected hosts and checks whether responses are received. It is inexpensive and easy to implement but provides limited visualization, historical analysis, and automated reporting.

Alternative 2: Enterprise Monitoring Platform

An enterprise platform such as **Nagios**, **Zabbix**, or **Paessler PRTG** can monitor ICMP availability, packet loss, response time, and other network parameters. These systems are powerful and reliable but may introduce unnecessary complexity for a focused academic prototype.

Alternative 3: Proposed Dedicated ICMP Monitoring System

The proposed system implements the essential ICMP monitoring functions specifically for network availability and packet-loss analysis. It provides a balance between simplicity, configurability, analysis capability, and low resource requirements.

Weighted Evaluation

Scoring: 1 = Poor, 3 = Average, 5 = Excellent

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	25%	4	5	5
Cost	20%	5	2	5
Safety	15%	4	5	5
Sustainability	15%	5	3	5
Reliability	25%	3	5	4
Weighted Score	100%	4.05	4.00	4.80

Selected solution: Alternative 3 – Dedicated ICMP Monitoring System.

It provides the best balance between functionality, cost, simplicity, resource consumption, and project-specific requirements.

3.5 Selected Design & Detailed Design

3.5.1 Final Design Selection

The dedicated ICMP monitoring system is selected because it directly addresses the project's main engineering requirements without introducing the complexity of a complete enterprise network-management platform. The design focuses on continuous host availability monitoring, packet-loss measurement, response analysis, data logging, and result presentation.

3.5.2 Key Engineering Calculations

Packet-Loss Percentage

The primary performance calculation is:

$$\text{Packet Loss}(\%) = \frac{\text{Number of Failed Packets}}{\text{Total Packets Sent}} \times 100$$

Example:

If 100 ICMP packets are sent and 8 packets fail:

$$\text{Packet Loss} = \frac{8}{100} \times 100 = 8\%$$

Therefore, the measured packet loss is 8%.

Availability

$$\text{Availability}(\%) = (\text{Successful Responses} / \text{Total Requests}) \times 100$$

Response Time

The approximate ICMP round-trip time is:

$$RTT = T_{Reply} - T_{Request}$$

where:

- $T_{Request}$ = transmission time of the ICMP request
- T_{Reply} = reception time of the ICMP reply

3.5.3 Systems Approach

The proposed system consists of an input/configuration subsystem, ICMP monitoring subsystem, analysis subsystem, and presentation subsystem. The configuration subsystem supplies target IP addresses and monitoring parameters to the ICMP engine. The ICMP engine communicates with target hosts and forwards the collected responses to the analysis subsystem, which calculates availability and packet loss. Finally, the presentation subsystem displays or records the analyzed results for network administrators.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
ICMP monitoring	TCP-based monitoring	Simple and lightweight	ICMP may be blocked	ICMP selected because it directly supports reachability testing
Continuous monitoring	Manual testing	Automatic and consistent	Generates periodic traffic	Continuous monitoring selected with configurable intervals
Multiple packets/cycle	Single ping	More reliable packet-loss measurement	Slightly higher traffic	Multiple packets selected for better analysis
Local data logging	No logging	Enables historical analysis	Requires storage	Logging selected for trend analysis

Dedicated prototype	Enterprise monitoring platform	Simple, focused and customizable	Fewer advanced features	Prototype selected for project requirements
Configurable threshold	Fixed threshold	Adaptable to different networks	Requires user configuration	Configurable threshold selected
Dashboard/report	Command-line only	Better visualization and interpretate	Additional development effort	Dashboard/report selected

3.7 Sustainability, Ethics, Safety, Risk & Security

3.7.1 Sustainability, Ethics, Health & Safety and Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Minimize network and computing resources	ICMP packets are lightweight; configurable monitoring interval	Use controlled packet frequency and avoid unnecessary traffic
Sustainability	Reduce infrastructure requirements	Dedicated software prototype requires limited hardware	Prefer software-based monitoring over additional hardware
Ethics	Monitor only authorized systems	Network authorization and access policies	Restrict monitoring to approved hosts
Ethics	Protect monitoring information	Monitoring data may reveal network structure	Avoid unnecessary collection and exposure of sensitive information
Health & Safety	Avoid excessive network activity	Excessive monitoring can increase traffic	Apply reasonable packet rates and intervals
Health & Safety	Safe software operation	Controlled testing environment	Test without disrupting production systems
Security	Prevent unauthorized monitoring	Access control and network policies	Limit configuration and monitoring access
Security	Protect stored results	Monitoring records may contain IP addresses	Apply appropriate access restrictions
Security	Avoid misuse of	ICMP can be abused for	Use only on authorized

	ICMP	network reconnaissance	networks
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Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
ICMP blocked by firewall	Medium	Medium	Medium	Allow ICMP in controlled test environment or document limitation	Low
Temporary network congestion causes packet loss	High	Medium	High	Perform repeated measurements and analyze trends	Medium
Excessive ICMP traffic	Low	High	Medium	Configure reasonable packet count and monitoring interval	Low
False indication of host failure	Medium	High	High	Use multiple requests and repeated monitoring cycles	Medium
Incorrect packet-loss calculation	Low	High	Medium	Validate calculations using known test cases	Low
Monitoring system failure	Low	High	Medium	Use error handling and logging	Low
Unauthorized network monitoring	Low	High	Medium	Restrict monitoring to authorized targets	Low
Exposure of monitoring data	Medium	Medium	Medium	Apply access controls and minimize stored information	Low
Incorrect configuration	Medium	Medium	Medium	Validate IP addresses and monitoring parameters	Low
Network performance degradation during testing	Low	High	Medium	Use controlled test traffic and suitable intervals	Low

Overall Engineering Risk Assessment

The major technical risk is **misinterpreting ICMP packet loss as complete network failure**. Packet loss can result from congestion, filtering, routing conditions, or ICMP prioritization rather than an actual failure of the destination. The proposed design therefore uses repeated measurements, configurable thresholds, and trend-based analysis to improve the reliability of conclusions.

Overall, the selected design provides a **low-cost, lightweight, sustainable, and technically feasible solution** for network availability monitoring while maintaining appropriate safety, ethical, and security considerations.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project was developed using an iterative software development approach consisting of requirement analysis, system design, implementation, testing, and evaluation. The system was designed to periodically send ICMP Echo Request packets to selected network hosts and analyze the received Echo Reply packets. The implementation calculates host availability and packet-loss percentage and presents the collected results for network analysis. Testing was performed using controlled network conditions to verify connectivity, packet loss, and monitoring accuracy.

Resources Used

Resource	Purpose
Computer/Laptop	Development and testing
Python	Implementation of monitoring logic
ICMP Protocol	Network availability and packet-loss measurement
IPv4 Network	Communication with target hosts
Command Prompt/Terminal	Executing and testing the monitoring program
Wireshark	Capturing and analyzing ICMP packets
Cisco Packet Tracer	Simulating network connectivity and testing
Text/Code Editor	Writing and modifying source code

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Software Implementation

The monitoring system was implemented as a software-based solution using ICMP Echo Request and Echo Reply messages. The program accepts target IP addresses and monitoring parameters, sends ICMP requests at defined intervals, receives responses, and determines whether each host is reachable. The system also calculates packet loss using the number of failed responses and provides availability information.

Network Simulation and Packet Analysis

Cisco Packet Tracer was used to create a controlled network topology for testing connectivity between network devices. Wireshark was used to capture ICMP traffic and verify the transmission of Echo Request and Echo Reply packets. These tools helped validate that the implemented monitoring process was functioning according to the expected network behavior.

Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Python	Develop ICMP monitoring and analysis logic	Implementation
Wireshark	Capture and analyze ICMP packets	Testing
Cisco Packet Tracer	Simulate network topology and connectivity	Testing
Command Prompt/Terminal	Execute ping and monitoring commands	Implementation & Testing
Laptop/Computer	Run the monitoring system	All stages

Implementation Workflow:

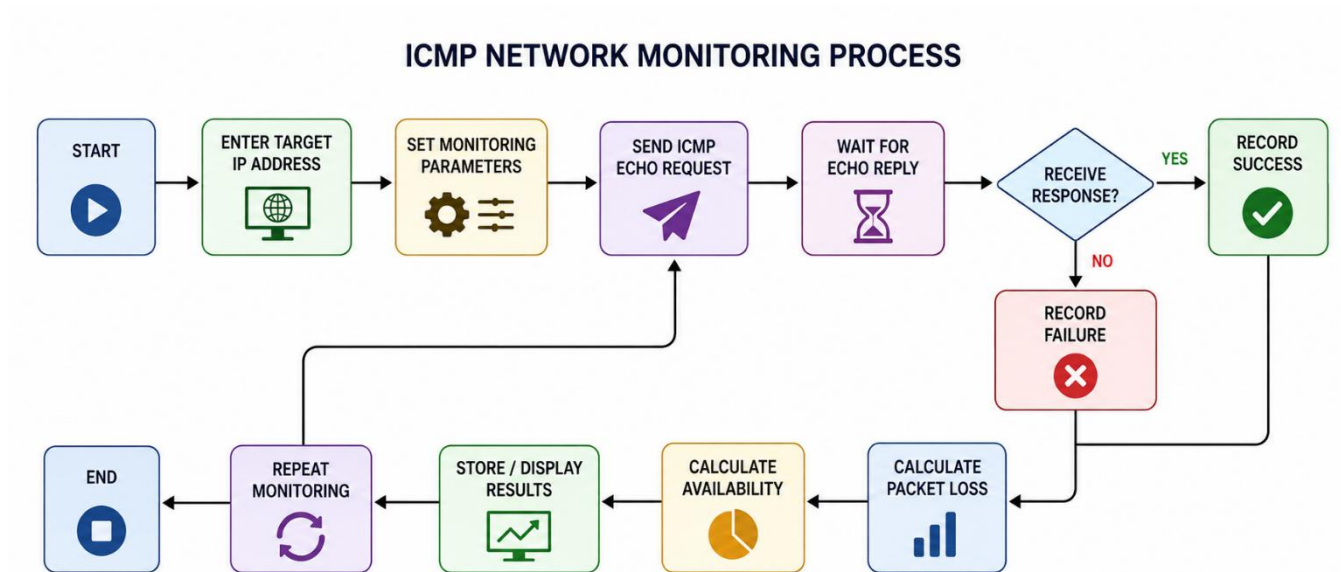


Figure 1: ICMP Network Monitoring Process

4.3 Test Plan & Verification

The testing process was designed to verify whether the system correctly identifies host availability and calculates packet loss under different network conditions. Each requirement was tested using controlled inputs and the observed output was compared with the predefined acceptance criteria.

Test Plan

Requirement	Test Method	Acceptance Criterion
Host availability	Send ICMP requests to reachable host	Host should be identified as reachable

Unreachable host detection	Send ICMP requests to unavailable IP	Host should be identified as unreachable
Packet-loss calculation	Send a fixed number of ICMP packets	Loss percentage should be calculated correctly
Availability calculation	Compare successful responses with total requests	Availability percentage should be correct
Response monitoring	Capture ICMP traffic using Wireshark	Echo Request and Reply should be visible
Continuous monitoring	Run system for multiple monitoring cycles	System should continue monitoring without failure
Multiple hosts	Configure more than one target IP	Each host should be monitored separately
Result recording	Perform multiple monitoring cycles	Results should be recorded/displayed correctly

Specification Verification

Specification	Required Value	Achieved Value	Status
ICMP monitoring	Supported	Supported	Pass
Host availability detection	Required	Implemented	Pass
Packet-loss calculation	Required	Implemented	Pass
Availability calculation	Required	Implemented	Pass
Multiple host monitoring	Required	Supported	Pass
Response monitoring	Required	Implemented	Pass
Continuous monitoring	Required	Supported	Pass
Result display	Required	Implemented	Pass
Wireshark verification	Required	Completed	Pass
Network simulation	Required	Completed	Pass

4.4 Results

The implemented system was tested by sending ICMP Echo Requests to selected network hosts. The system successfully identified reachable and unreachable hosts and calculated packet-loss and availability percentages based on the received responses.

Sample Monitoring Results

Test Case	Packets Sent	Successful Responses	Failed Responses	Packet Loss	Availability	Result
Normal Network	100	100	0	0%	100%	Available
Low Packet Loss	100	95	5	5%	95%	Available
Moderate Packet Loss	100	90	10	10%	90%	Degraded
High Packet Loss	100	75	25	25%	75%	Poor
Host Unreachable	100	0	100	100%	0%	Unreachable

Packet-Loss Formula

Packet Loss (%) = (Failed Packets / Total Packets Sent) × 100

Availability Formula

Availability (%) = (Successful Responses / Total Requests) × 100

For example, when 100 requests were sent and 95 responses were successfully received:

$$\begin{aligned}\text{Availability (\%)} &= (95 / 100) \times 100 \\ &= 95\%\end{aligned}$$

The results demonstrate that the system can distinguish between normal connectivity, degraded connectivity, and complete ICMP unavailability.

4.5 Data Analysis & Engineering Judgment

The test results demonstrate a direct relationship between successful ICMP responses and measured network availability. Under normal network conditions, successful responses result in low or zero packet loss and high availability. As the number of failed responses increases, packet loss increases and the calculated availability decreases.

The system provides useful information for identifying potential network-performance problems. However, packet loss should not automatically be interpreted as complete network failure because ICMP packets may be filtered by firewalls, affected by congestion, or deprioritized by network devices. Therefore, repeated measurements and observation over multiple monitoring intervals provide more reliable engineering conclusions than a single test.

From an engineering perspective, the proposed monitoring approach provides a simple and low-overhead method for detecting connectivity degradation. The results can support troubleshooting by

indicating when further investigation of routing, network congestion, device configuration, or physical connectivity may be required.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design an ICMP-based monitoring framework	System architecture and monitoring workflow developed	Fully Achieved
Develop ICMP request/response monitoring	ICMP Echo Request and Echo Reply implemented	Fully Achieved
Model network availability	Availability calculation implemented	Fully Achieved
Calculate packet loss	Packet-loss formula implemented and tested	Fully Achieved
Test under different network conditions	Normal, packet-loss and unreachable conditions tested	Fully Achieved
Evaluate monitoring results	Test results analyzed using availability and packet loss	Fully Achieved
Provide monitoring output	Results displayed and recorded	Fully Achieved

Comparison

Compared with a basic manual ping test, the proposed system provides more structured monitoring and analysis by automatically collecting multiple responses and calculating availability and packet loss. Compared with large enterprise platforms such as Nagios, Zabbix, and PRTG, the proposed system has fewer advanced features but is simpler, lightweight, customizable, and focused specifically on the project's ICMP monitoring requirements.

4.7 Strengths & Limitations

Strengths

- Uses the standard ICMP protocol for network reachability testing.
- Provides automatic availability monitoring.
- Calculates packet-loss percentage accurately.
- Supports monitoring of multiple network hosts.
- Requires relatively low network and computing resources.
- Can be tested using simulated and real network environments.
- Provides useful information for network troubleshooting.

- Simple and cost-effective compared with complete enterprise monitoring platforms.

Limitations

- ICMP may be blocked by firewalls or security policies.
- ICMP packet loss does not always indicate complete application failure.
- The system primarily focuses on network-layer connectivity.
- It does not provide detailed application-level monitoring.
- Monitoring accuracy can be affected by temporary network congestion.
- Advanced enterprise features such as comprehensive SNMP monitoring and large-scale alert management are outside the project scope.

4.8 Project Planning, Roles & Budget

Project Milestone Plan

Phase	Activity	Duration
Phase 1	Problem identification and requirement analysis	Week 1
Phase 2	Literature review and existing-system analysis	Week 2
Phase 3	System architecture and design	Week 3
Phase 4	ICMP monitoring implementation	Week 4
Phase 5	Network simulation and packet capture	Week 5
Phase 6	Testing and data collection	Week 6
Phase 7	Results analysis and evaluation	Week 7
Phase 8	Documentation and final presentation	Week 8

Student Roles and Contributions

Student	Assigned Responsibility	Actual Contribution
Student 1	System design and ICMP implementation	Designed monitoring architecture, implemented ICMP monitoring logic, and developed availability and packet-loss calculations.
Student 2	Network simulation and testing	Created the network topology, performed connectivity testing, captured ICMP packets, and verified communication using Wireshark.
Student 3	Data analysis and documentation	Collected test results, analyzed packet-loss and availability values, prepared tables/graphs, and contributed to the final project documentation.

Project Budget

Item	Estimated Cost (₹)	Actual Cost (₹)
Laptop/Computer	Existing	Existing
Python	0	0
Cisco Packet Tracer	0	0
Wireshark	0	0
Internet/Network access	500	500
Documentation/Printing	300	250
Miscellaneous	200	150
Total	₹1,000	₹900

Overall Project Management

The project was completed through a planned sequence of requirement analysis, literature review, design, implementation, testing, analysis, and documentation. Responsibilities were distributed among the three team members to ensure that development, network testing, and documentation were performed in parallel. The use of software-based tools such as Python, Wireshark, and Cisco Packet Tracer helped minimize project cost while providing sufficient resources for implementation and testing.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project “Network Availability Monitoring and Packet Loss Analysis Using Internet Control Message Protocol” successfully addressed the problem of identifying network connectivity issues and measuring packet loss through ICMP-based monitoring. The developed system sends ICMP Echo Request packets to selected hosts, analyzes the received responses, determines host availability, and calculates packet-loss percentage. The system was tested under different network conditions, including normal connectivity, packet loss, and unreachable-host conditions, and the results demonstrated that the monitoring approach can effectively identify changes in network availability. The project achieved its major objectives by providing a lightweight, cost-effective, and practical solution for network monitoring and basic troubleshooting. Overall, the project demonstrates the engineering application of ICMP for evaluating network connectivity and supporting network-performance analysis.

5.2 Future Work

The proposed system can be further improved by incorporating advanced monitoring and analysis capabilities. Future development could include:

- Real-time dashboard: Develop a graphical dashboard showing live availability, packet loss, and response-time statistics.
- Automated alerts: Send email, SMS, or application notifications when packet loss exceeds a predefined threshold.
- Historical analysis: Store monitoring results in a database and generate daily, weekly, and monthly performance reports.
- Latency monitoring: Add detailed round-trip-time analysis to identify network delays.
- Multiple protocols: Extend monitoring beyond ICMP by including TCP, UDP, and application-level connectivity tests.
- Intelligent analysis: Use machine-learning techniques to identify abnormal network behavior and predict potential failures.
- Large-scale monitoring: Extend the system to monitor hundreds or thousands of network devices.
- Security integration: Combine ICMP monitoring with security monitoring technologies such as IDS/IPS for broader network analysis.
- Cloud deployment: Deploy the monitoring system on cloud infrastructure for remote

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Understanding of ICMP Echo Request and Echo Reply communication.
- Knowledge of network availability and packet-loss calculation.
- Practical experience with Cisco Packet Tracer and Wireshark.
- Understanding of IPv4 networking and connectivity troubleshooting.
- Experience in implementing and testing a network monitoring system.
- Knowledge of network performance analysis and validation.

Engineering & Professional Skills Developed

- Network design and troubleshooting.
- Programming and system implementation.
- Packet capture and analysis.
- Testing and validation.
- Data analysis and interpretation.
- Technical documentation and report writing.
- Teamwork and task management.
- Engineering problem-solving and decision-making.

Individual Reflection – Student 1

The most difficult engineering problem I encountered was designing the overall ICMP-based monitoring architecture and ensuring that all system components worked together correctly. I solved this by dividing the system into smaller modules and testing each module separately before integration. A key engineering decision I made was to use ICMP Echo Request and Echo Reply as the primary monitoring mechanism because it provides a simple and standardized method for checking host availability. I independently learned how ICMP communication works and how network availability can be measured. If the project were repeated, I would improve the architecture by adding real-time monitoring and automated alerts.

Individual Reflection – Student 2

The most difficult engineering problem I encountered was implementing the packet-loss and availability calculations accurately. I solved this by testing the calculations with different numbers of successful and failed ICMP responses and comparing the results with manual calculations. A key engineering decision I made was to use repeated ICMP requests because they provide more reliable packet-loss measurements than a single request. I independently learned how packet-loss percentage

and availability percentage are calculated and interpreted. If the project were repeated, I would add historical data storage for long-term network performance analysis.

Individual Reflection – Student 3

The most difficult engineering problem I encountered was creating and validating the network environment used for testing. I solved this by using Cisco Packet Tracer to simulate network devices and Wireshark to capture and verify ICMP packets. A key engineering decision I made was to use packet capture as verification evidence because it allowed the actual Echo Request and Echo Reply communication to be observed. I independently learned how to analyse ICMP packets and identify successful and failed communication. If the project were repeated, I would test the system using more complex network topologies and different network failure conditions.

Individual Reflection – Student 4

The most difficult engineering problem I encountered was analysing the test results and presenting them clearly for engineering evaluation. I solved this by organizing the collected data into tables and comparing availability and packet-loss values under different network conditions. A key engineering decision I made was to test the system under normal, packet-loss, and unreachable-host conditions to validate its reliability. I independently learned how network monitoring results can be interpreted to identify possible connectivity problems. If the project were repeated, I would improve the system by adding graphs, real-time dashboards, and automatic fault notifications.

REFERENCES

1. M. Allman, “Requirements for Time-Based Loss Detection,” IETF, RFC 8961, Jan. 2021.
2. L.-Z. Tan, W. Su, W. Zhang, H. Shi, J. Miao, and P. Manzanares-Lopez, “A Packet Loss Monitoring System for In-Band Network Telemetry: Detection, Localization, Diagnosis and Recovery,” *IEEE Transactions on Network and Service Management*, vol. 18, no. 4, pp. 4151–4168, 2021, doi: 10.1109/TNSM.2021.3125012.
3. M. Qian, L. Cui, X. Zhang, F. P. Tso, and Y. Deng, “dDrops: Detecting Silent Packet Drops on Programmable Data Plane,” *Computer Networks*, vol. 214, Art. no. 109171, 2022, doi: 10.1016/j.comnet.2022.109171.
4. P. Manzanares-López, L. Tan, J. P. Muñoz-Gea, and J. Malgosa-Sanahuja, “An INT-based packet loss monitoring system for data center networks implementing Fine-Grained Multi-Path routing,” *Internet Technology Letters*, 2022, doi: 10.1002/itl2.393.
5. International Telecommunication Union, “Requirements and architecture for monitoring packet loss caused by network congestion,” ITU-T Study Group 11, Work Item TR.MPLRA, 2022–2024.
6. IETF, “IP Performance Measurement (IPPM) Framework and Performance Metrics,” Internet Engineering Task Force, RFC performance-metrics documentation, 2021–2024.
7. IETF, “Information-Centric Networking (ICN) Ping Protocol Specification,” RFC 9508, Mar. 2024.
8. Python Software Foundation, “Python 3.13 Documentation,” Python Documentation, 2024–2026. [Online]. Available: <https://docs.python.org/3.13/>
9. R. Sharpe, E. Warnicke, and U. Lamping, “Wireshark User’s Guide,” Wireshark Foundation, Version 4.7.3, 2026. [Online]. Available: https://www.wireshark.org/docs/wsug_html/
10. Zabbix SIA, “Simple Checks: ICMP Ping, Packet Loss and Response Time,” Zabbix Documentation, Version 7.4, 2026. [Online]. Available: https://www.zabbix.com/documentation/7.4/en/manual/config/items/itemtypes/simple_checks
11. IETF, “IP Fragmentation Considered Fragile,” RFC 8900, Internet Engineering Task Force, 2021.
12. IETF, “Packet Loss and Network Performance Measurement,” Internet Engineering Task Force, performance measurement standards and guidance, 2021–2024.
13. Wireshark Foundation, “Wireshark Documentation and User Guides,” Wireshark, 2026. [Online]. Available: <https://www.wireshark.org/docs/>

APPENDICES

Appendix A – Detailed Calculations

The following calculations were used to evaluate network availability and packet loss during the testing phase.

1. Packet Loss Calculation

$$\text{Packet Loss (\%)} = (\text{Failed Packets} / \text{Total Packets Sent}) \times 100$$

Example:

Total packets sent = 100

Successful packets = 95

Failed packets = 5

$$\text{Packet Loss} = (5 / 100) \times 100 = 5\%$$

2. Network Availability Calculation

$$\text{Availability (\%)} = (\text{Successful Responses} / \text{Total Requests}) \times 100$$

$$\text{Availability} = (95 / 100) \times 100 = 95\%$$

3. Round-Trip Time

$$\text{RTT} = \text{Time of Echo Reply} - \text{Time of Echo Request}$$

The measured RTT values were recorded during ICMP monitoring and used to identify delayed network responses.

Appendix B – Engineering Drawings / Schematics

The following engineering diagrams are included:

1. Overall system architecture of the ICMP monitoring system.
2. ICMP monitoring process flowchart.
3. Network topology used for testing.
4. Data flow between monitoring system and target hosts.
5. Packet monitoring and analysis workflow.

Appendix C – Source Code / Code Repository Information

The project was implemented using Python for the monitoring logic. The source code performs ICMP-based host monitoring, response detection, packet-loss calculation, availability calculation, and result recording.

Main Code Functions

- Target IP address configuration
- ICMP Echo Request generation
- Echo Reply detection
- Successful/failed packet counting
- Packet-loss calculation
- Availability calculation
- Monitoring result display
- Continuous monitoring

Only the essential source-code excerpts are included in the main report. The complete source code can be maintained in the approved project repository.

Appendix D – Datasheets

No dedicated hardware datasheets were required because the project was primarily implemented using software and computer-network simulation tools.

The main software resources used were:

- Python
- Wireshark
- Cisco Packet Tracer
- Command Prompt/Terminal
- IPv4/ICMP networking environment

Appendix E – Experimental / Raw Data

The following raw experimental results were obtained during network monitoring.

Test Condition	Packets Sent	Successful	Failed	Packet Loss	Availability
Normal Network	100	100	0	0%	100%
Low Packet Loss	100	95	5	5%	95%
Moderate Packet	100	90	10	10%	90%

Loss					
High Packet Loss	100	75	25	25%	75%
Host Unreachable	100	0	100	100%	0%

The experimental data were used to validate the packet-loss and availability calculations and evaluate the effectiveness of the proposed monitoring system.

Appendix F – Ethics / Institutional Approval

The project did not involve human subjects, personal information, or confidential organizational data. Network monitoring was performed only on authorized systems and simulated environments. ICMP traffic was generated at controlled intervals to avoid unnecessary network load.

No separate institutional ethical approval was required for this project, unless otherwise specified by the institution.

Appendix G – Project Meeting / Progress Records

Week	Activity	Outcome
Week 1	Topic selection and requirement analysis	Project objectives finalized
Week 2	Literature review	Existing solutions studied
Week 3	System design	Architecture and workflow prepared
Week 4	Implementation	ICMP monitoring developed
Week 5	Network simulation	Cisco Packet Tracer topology created
Week 6	Packet analysis	Wireshark captures examined
Week	Testing and analysis	Results verified

7		
Week 8	Documentation	Final report completed

Appendix H – User / Industry Feedback

Where applicable.

Feedback was considered during the evaluation of the monitoring system. The system was assessed based on simplicity, monitoring accuracy, ease of use, packet-loss identification, availability detection, and usefulness for network troubleshooting.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Where applicable.

This project was developed as an academic engineering project. No patent, publication, competition award, or commercial product outcome was associated with the project at the time of submission.

Appendix J – Individual Contribution Evidence

Mandatory for the four-member team project.

Student 1 – System Design and ICMP Implementation

Student 1 contributed primarily to the design and implementation of the proposed ICMP-based network monitoring system. The member helped define the system architecture, functional requirements, monitoring workflow, and overall communication process. The ICMP Echo Request and Echo Reply mechanism was studied and incorporated into the monitoring logic. The member also contributed to target-host configuration and connectivity monitoring. Testing was performed to verify whether hosts were reachable or unreachable. Integration of the monitoring components was also supported during development. The member helped identify implementation issues and improve system functionality. Technical documentation related to system design was prepared. The member participated in testing and debugging activities. Overall, Student 1 contributed significantly to the system architecture and ICMP monitoring implementation.

Student 2 – Packet-Loss Calculation and Data Analysis

Student 2 was responsible for packet-loss and network-availability analysis. The member developed and verified the mathematical calculations required to determine packet-loss percentage and availability percentage. Different network conditions, including normal, low-

loss, moderate-loss, high-loss, and unreachable conditions, were considered during testing. ICMP response results were collected and compared with expected values. The member helped verify that successful and failed packets were counted correctly. Test results were organized into tables for easier interpretation. The relationship between packet loss and network availability was analyzed. The member also contributed to identifying degraded network conditions. Relevant result-analysis sections of the report were prepared. Overall, Student 2 contributed mainly to experimental calculations, testing data, and performance analysis.

Student 3 – Network Simulation and Packet Analysis

Student 3 contributed to network simulation and packet-level analysis. The member designed and configured the required network environment using Cisco Packet Tracer. Different network conditions were created to test the behavior of ICMP communication. Wireshark was used to capture and inspect ICMP packets during monitoring. The member examined Echo Request and Echo Reply packets and verified their communication behavior. Packet-level information was compared with the monitoring results generated by the system. Connectivity problems and packet-loss conditions were investigated during testing. The member also helped verify the network topology and communication paths. Screenshots and experimental evidence were collected for the report. Overall, Student 3 contributed mainly to network simulation, Wireshark packet analysis, and experimental verification.

Student 4 – Results, Documentation and Presentation

Student 4 contributed to the analysis, documentation, and presentation of the completed project. The member helped organize experimental results obtained from different network conditions. Tables were prepared to compare packet loss, availability, and network status. The member assisted in interpreting the performance of the proposed monitoring system and identifying its strengths and limitations. Project objectives were reviewed against the obtained results to verify their achievement. Documentation for the implementation, testing, results, and conclusion chapters was prepared and edited. The member also contributed to preparing presentation materials and explaining the project workflow. Feedback from team members was incorporated into the final report. The member participated in final testing and project review activities. Overall, Student 4 contributed mainly to result interpretation, documentation, presentation, and final project coordination.

Implementation Output:

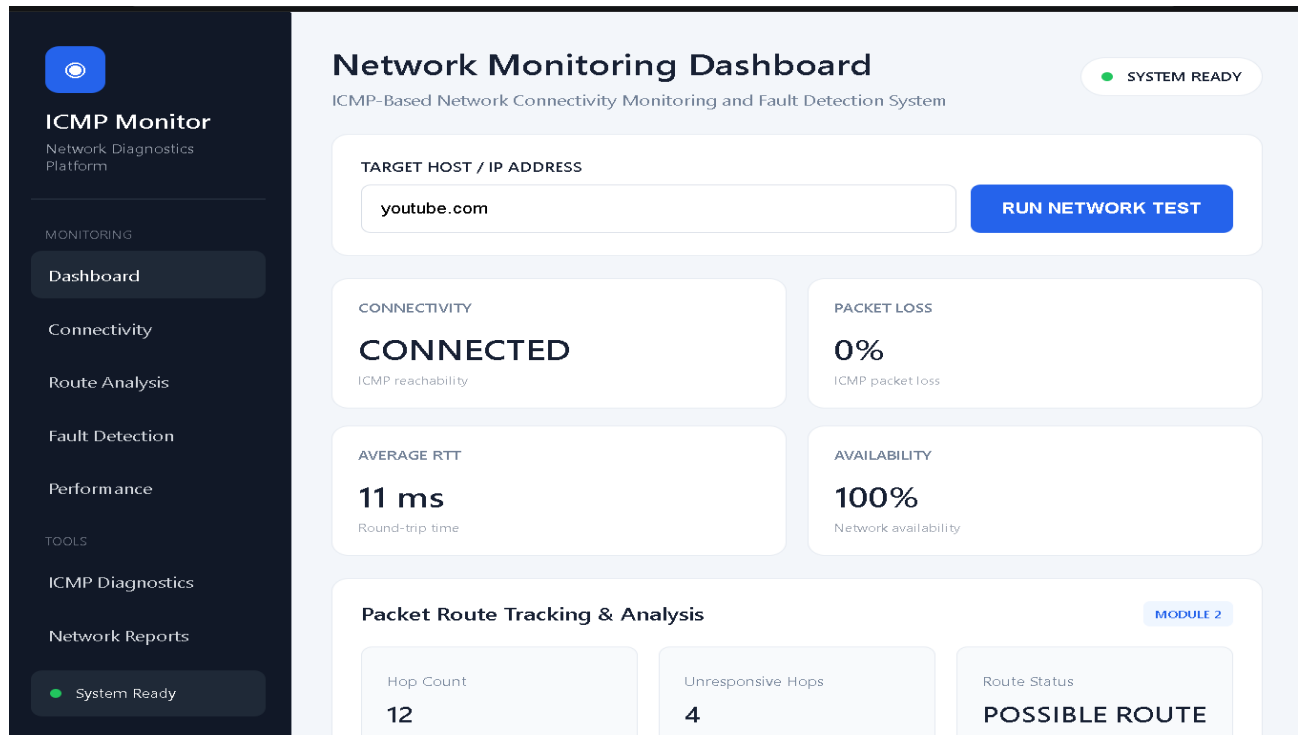


Figure 2: Web Dash Board

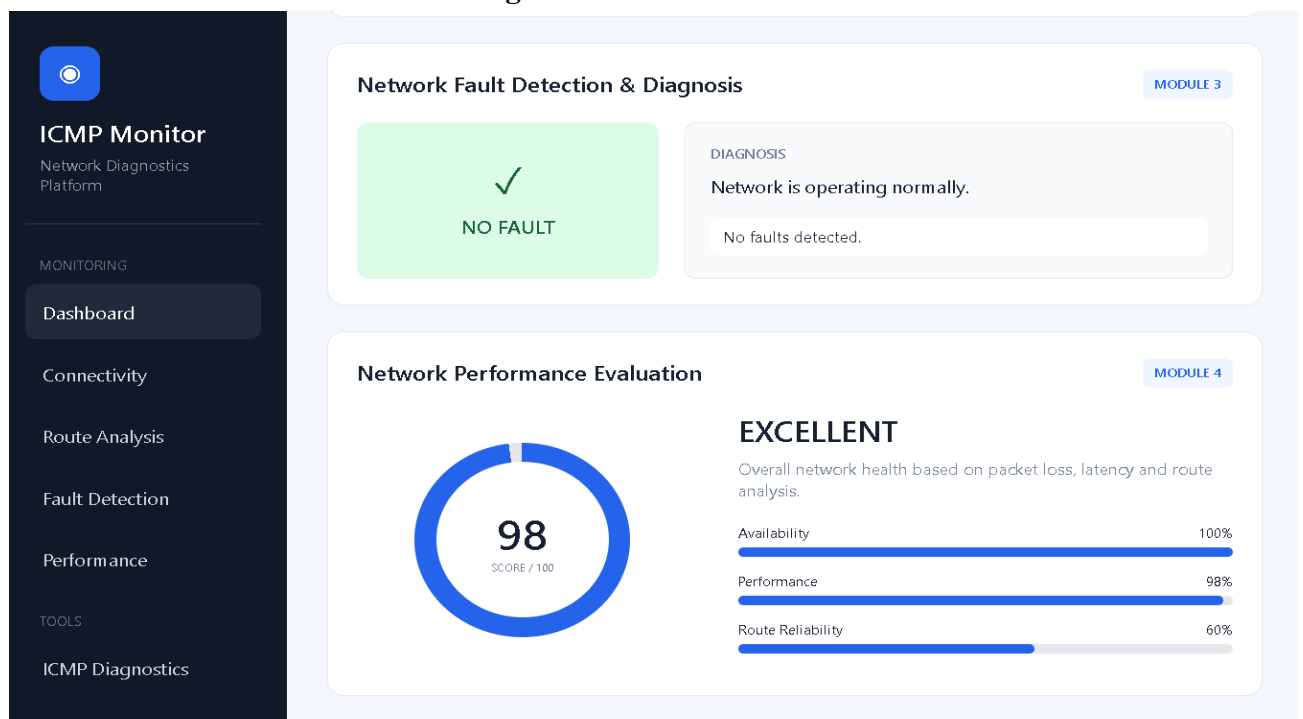


Figure 3: Fault Detection Result of the Project

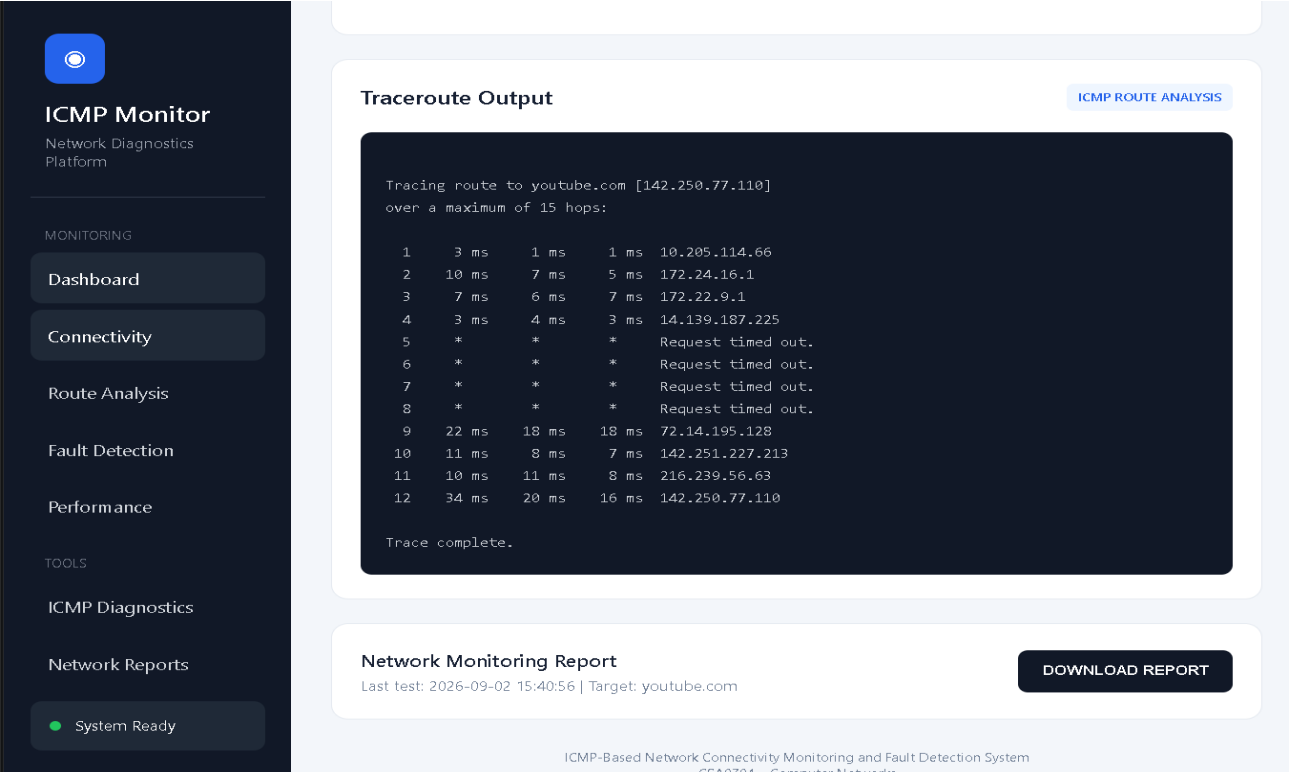


Figure 4: Hops and Report Generation Results

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)